Dual-Domain MMD Adaptation: Regularized Re-Run Update

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1 What changed

The synthetic→real adaptation direction (`syn_source_real_target`) was re-run with two additional regularization mechanisms enabled, keeping everything else (data, architecture, epochs, layer, kernel) identical to the original run:

- **Linear MMD-weight decay**: λ_{DA} now decays linearly from 1.0 to 0.1 over the course of training, instead of staying constant at 0.8 throughout.
- **Bandwidth freeze at epoch 5**: the RBF kernel's bandwidth stops updating after epoch 5 and is held fixed for the remainder of training, instead of continuing to track the batch statistics via exponential moving average for all 100 epochs.

2 Result

Model	Eval domain	mAP50	mAP50-95
baseline (syn-pretrained)	syn (in-domain)	0.758	0.508
baseline (real-pretrained)	real (in-domain)	0.896	0.641
syn_source_real_target (original)	real (target)	0.867	0.599
syn_source_real_target (original)	syn (source)	0.568	0.337
syn_source_real_target_reg (new)	**real (target)**	**0.886**	**0.618**
syn_source_real_target_reg (new)	**syn (source)**	**0.674**	**0.396**

Table 1: Original vs. regularized re-run of the synthetic→real direction, evaluated on the same stride-3 subsampled validation sets used for the pretrain baselines.

The regularized re-run improves on both fronts simultaneously: target-domain (real) mAP50 rises from 0.867 to 0.886, closing most of the remaining gap to the in-domain baseline (0.896). More notably, source-domain (syn) performance – the forgetting metric – recovers substantially, from 0.568 to 0.674 mAP50, more than halving the regression relative to the syn baseline (0.758).

3 The EMA bandwidth mechanism

The RBF kernel's bandwidth is not fixed; it is estimated as an exponential moving average of the batch's mean pairwise squared distance, updated every training step. As the two domains' features grow closer together under MMD alignment, this estimate shrinks too, which sharpens the kernel and rewards the optimizer for collapsing the representation further rather than genuinely matching the two distributions – a feedback loop that self-reinforces as training progresses. Freezing the bandwidth after a short warm-up period (epoch 5, by which point the EMA has already smoothed over hundreds of batches) breaks this loop while still using a representative early-training estimate, which is the most likely explanation for the reduced forgetting observed above.